# Expectation, Variance, and Discrete Distributions

50 minutes • 50 points • Instructor key

**Grading note.** Award method credit for correct reasoning after an earlier error; avoid repeated deductions for the same arithmetic mistake. Suggested working time is 46 minutes, leaving 4 minutes for checking.

**Instructions.** Answer all four problems and show your reasoning. No calculator is needed; exact fractions or exact expressions with powers and binomial coefficients are acceptable. You may use the standard binomial and geometric mean and variance formulas without proof. Geometric waiting times count trials *including the first success*. The suggested times leave 4 minutes to check your work.

## 1. From outcomes to moments

**14 points** · about 13 minutes

Let $\Omega = \{a, b, c, d\}$, with every subset an event. The outcome probabilities and the values of a random variable $X$ are given below; $q$ is a constant.

| Outcome $\omega$ | $a$ | $b$ | $c$ | $d$ |
|---|---|---|---|---|
| $\mathbb{P}(\{\omega\})$ | $q$ | $2q$ | $q$ | $4q$ |
| $X(\omega)$ | $-2$ | $0$ | $2$ | $2$ |

(a) Find $q$. List the outcomes in the event $\{X > 0\}$ and find $\mathbb{P}(X > 0)$. (3 points)

**Solution.** The total mass is $8q = 1$, so $q = 1/8$; all masses are nonnegative. The event is $\{c, d\}$ and its probability is $q + 4q = 5/8$.

*Credit:* Normalization and $q$: 1; event: 1; probability: 1.

(b) Give the probability mass function of $X$: list each distinct possible value and its probability. (3 points)

**Solution.** $\mathbb{P}(X = -2) = 1/8$, $\mathbb{P}(X = 0) = 1/4$, and $\mathbb{P}(X = 2) = 5/8$, with probability zero at all other values. The outcomes $c, d$ contribute to the same value, so their masses add.

*Credit:* Each value with its correct probability: 1.

(c) Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\mathrm{Var}(X)$. (5 points)

**Solution.**

$$\mathbb{E}[X] = (-2)\frac{1}{8} + 0\frac{1}{4} + 2\frac{5}{8} = 1, \qquad \mathbb{E}[X^2] = 4\frac{1}{8} + 0\frac{1}{4} + 4\frac{5}{8} = 3.$$

Hence $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 3 - 1^2 = 2$.

*Credit:* Weighted first moment and value: 2; second moment: 1; variance identity: 1; variance value: 1.

(d) Let $Y = 3 - 2X$. Find $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$, stating the transformation rules you use. (3 points)

**Solution.** $\mathbb{E}[Y] = 3 - 2\mathbb{E}[X] = 1$ and $\mathrm{Var}(Y) = (-2)^2 \mathrm{Var}(X) = 4 \cdot 2 = 8$. Adding a constant leaves variance unchanged; scaling squares the factor.

*Credit:* Mean with linearity: 1; squared scaling rule: 1; variance value: 1. A direct distribution calculation also earns full credit.

## 2. Bernoulli indicators

**10 points** · about 9 minutes

Roll a fair six-sided die once. Let $I = 1$ if the result is 5 or 6, and let $I = 0$ otherwise.

(a) State the distribution of $I$, including its parameter, and give $\mathbb{P}(I = 0)$ and $\mathbb{P}(I = 1)$. (2 points)

**Solution.** $I \sim \text{Bernoulli}(1/3)$, with $\mathbb{P}(I = 0) = 4/6 = 2/3$ and $\mathbb{P}(I = 1) = 2/6 = 1/3$.

*Credit:* Bernoulli family and parameter: 1; both masses: 1.

(b) Use the two possible values of $I$ to calculate $\mathbb{E}[I]$, $\mathbb{E}[I^2]$, and $\text{Var}(I)$. Show the weighted sums rather than only quoting a Bernoulli formula. (4 points)

**Solution.**

$$\mathbb{E}[I] = 0 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} = \frac{1}{3}, \qquad \mathbb{E}[I^2] = 0^2 \cdot \frac{2}{3} + 1^2 \cdot \frac{1}{3} = \frac{1}{3}.$$

Therefore $\text{Var}(I) = 1/3 - (1/3)^2 = 2/9$. The identity $I^2 = I$ also explains why the first two moments agree.

*Credit:* First-moment sum: 1; second-moment sum or $I^2 = I$: 1; variance identity: 1; value: 1.

(c) On the *same roll*, let $J = 1 - I$. Find $\mathbb{E}[I + J]$ and $\text{Var}(I + J)$. A student claims $\text{Var}(I + J) = \text{Var}(I) + \text{Var}(J) = 4/9$. Explain why this reasoning fails here. (4 points)

**Solution.** Since $I + J = 1$ on every outcome, $\mathbb{E}[I + J] = 1$ and $\text{Var}(I + J) = 0$. Linearity of expectation is valid without independence. However, $I$ and $J$ are dependent: knowing either determines the other. Thus the independent-variance addition rule does not apply. For example, $\mathbb{P}(I = 1, J = 1) = 0$, whereas $\mathbb{P}(I = 1)\mathbb{P}(J = 1) = (1/3)(2/3) = 2/9$.

*Credit:* Mean: 1; variance: 1; identify dependence and explain why it invalidates this use of variance addition: 2. A covariance calculation is not required.

## 3. Binomial counts

**12 points** · about 11 minutes

A bag contains one red ball and three blue balls. Draw one ball uniformly at random, record its color, and replace it. Repeat for a total of six draws. Assume the draws are independent. Let $N$ be the number of red balls recorded.

(a) State the distribution of $N$, its parameters, and its possible values. Explain why the model applies. (3 points)

**Solution.** $N \sim \text{Bin}(6, 1/4)$, supported on $0, 1, \ldots, 6$. There are six fixed trials, each records red or not red, the red probability stays $1/4$ because of replacement, and the draws are independent by assumption.

*Credit:* Family and parameters: 1; support: 1; modeling assumptions: 1.

(b) Find $\mathbb{P}(N = 2)$. Explain the counting factor in your expression. (3 points)

**Solution.**

$$\mathbb{P}(N = 2) = \binom{6}{2}\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^4 = \frac{1215}{4096}.$$

The factor $\binom{6}{2} = 15$ chooses the two red positions among the six draws. Each such red/blue sequence has the same probability.

*Credit:* Correct probability of a specified sequence: 1; binomial factor: 1; explanation of that factor: 1.

(c) Find the probability that at least one red ball is recorded. (2 points)

**Solution.**

$$\mathbb{P}(N \geq 1) = 1 - \mathbb{P}(N = 0) = 1 - \left(\frac{3}{4}\right)^6 = \frac{3367}{4096}.$$

*Credit:* Complement or equivalent correct sum: 1; correct probability expression: 1.

(d) Compute $\mathbb{E}[N]$ and $\text{Var}(N)$. If $N$ is written as a sum of six success indicators, why may their variances be added? (4 points)

**Solution.**

$$\mathbb{E}[N] = 6 \cdot \frac{1}{4} = \frac{3}{2}, \qquad \text{Var}(N) = 6 \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{9}{8}.$$

Each indicator records one of the independent draws, so the indicators are independent and their variances add. Their expectations add by linearity, which itself does not require independence.

*Credit:* Mean: 1; variance formula: 1; variance value: 1; independence justification: 1.

## 4. Geometric waiting times

**14 points** · about 13 minutes

A device attempts to send a message repeatedly. Each attempt succeeds with probability 1/4, independently of all other attempts. Let $T$ be the number of attempts *up to and including the first success.*

(a) Give the possible values of $T$ and a formula for $\mathbb{P}(T = k)$. Explain the formula and use a geometric series to check that its probabilities sum to 1. (4 points)

**Solution.** The support is $1, 2, 3, \ldots$, and $\mathbb{P}(T = k) = (3/4)^{k-1}(1/4)$: the first $k - 1$ attempts fail and the $k$th succeeds. Independence lets us multiply these probabilities.

$$\sum_{k=1}^{\infty} \mathbb{P}(T = k) = \frac{1}{4} \sum_{j=0}^{\infty} \left(\frac{3}{4}\right)^{j} = \frac{1/4}{1 - 3/4} = 1.$$

*Credit:* Support: 1; PMF: 1; failure/success explanation: 1; normalization: 1.

(b) Find $\mathbb{P}(T = 3)$ and $\mathbb{P}(T > 3)$. (2 points)

**Solution.** $\mathbb{P}(T = 3) = (3/4)^2(1/4) = 9/64$. The event $T > 3$ means that the first three attempts fail, so $\mathbb{P}(T > 3) = (3/4)^3 = 27/64$.

*Credit:* Each correct probability: 1.

(c) Find $\mathbb{E}[T]$ and $\mathrm{Var}(T)$. (3 points)

**Solution.** With $p = 1/4$ and trials counted through the first success,

$$\mathbb{E}[T] = \frac{1}{p} = 4, \qquad \mathrm{Var}(T) = \frac{1 - p}{p^2} = \frac{3/4}{(1/4)^2} = 12.$$

*Credit:* Mean: 1; variance formula: 1; variance value: 1.

(d) Let $F$ be the number of *failed* attempts before the first success. Express $F$ in terms of $T$, then find $\mathbb{E}[F]$ and $\mathrm{Var}(F)$. (2 points)

**Solution.** $F = T - 1$, so $\mathbb{E}[F] = \mathbb{E}[T] - 1 = 3$ and $\mathrm{Var}(F) = \mathrm{Var}(T) = 12$; subtracting a constant does not change variance.

*Credit:* Relation and mean: 1; unchanged variance: 1.

(e) Given that the first two attempts have failed, find the probability that more than five attempts in total will be needed: $\mathbb{P}(T > 5 \mid T > 2)$. Justify your answer. (3 points)

**Solution.** Since $\{T > 5\} \subseteq \{T > 2\}$,

$$\mathbb{P}(T > 5 \mid T > 2) = \frac{\mathbb{P}(T > 5)}{\mathbb{P}(T > 2)} = \frac{(3/4)^5}{(3/4)^2} = \left(\frac{3}{4}\right)^3 = \frac{27}{64}.$$

Equivalently, attempts 3, 4, and 5 must all fail; independence keeps their conditional failure probability at 3/4 each. This is the geometric memoryless property.

*Credit:* Conditional-probability setup or identify three further failures: 1; justified cancellation or independence: 1; result: 1.